● Critical — must have ● High — important ● Medium — adds value ● Low — nice to have

PHASE 0 — DATA FOUNDATION

Synthetic influencer profiles
Critical

Generate 50 realistic profiles across 8 niches with follower count, engagement rate, demographics, and content style. Used for all MVP testing and demos.

Synthetic brand profiles
Critical

Generate 20 brand profiles with product niche, target audience, budget range, and collaboration preference. Seeds the matching engine from day one.

Realistic data distributions
High

Engagement rate inversely correlated with follower count plus noise, follower counts drawn from realistic log-normal distribution, not random flat values.

Database seeding script
Critical

Python script that populates all tables from the synthetic dataset. Idempotent — can be re-run to reset the database to a clean demo state.

PHASE 1 — DATABASE & CORE DATA MODEL

Influencers table
Critical

ID, name, niche, follower count, engagement rate, location, content format tags, bio. Indexed on niche, engagement rate, location for fast query performance.

Brands table
Critical

ID, name, industry, target audience description, budget range, preferred niche. Linked to matches table via foreign key.

Past campaigns table
Critical

Stores 2–3 historical collaboration records per influencer. Brand category, campaign type, estimated reach, outcome tag. Used in history-fit sub-score.

Matches table
Critical

Stores every brand–influencer pairing with total match score and all four sub-scores. Allows re-querying without recomputing. Indexed on brand ID.

Niche similarity matrix
High

Lookup table defining adjacent niche pairs and their similarity score (e.g. fitness ↔ wellness = 75, fitness ↔ gaming = 10). Powers niche alignment sub-score.

Engagement tier benchmarks
High

Defines expected engagement rate per follower tier (5K–20K, 20K–50K, 50K–100K). Used to normalize engagement quality score against realistic baselines.

PHASE 2 — MATCHING ALGORITHM

Niche alignment scorer
Critical

Compares brand category to influencer niche and past campaign categories. Returns 0–100 using exact match, similarity matrix, or mismatch logic. Weight: 35%.

Audience compatibility scorer
Critical

Scores age group, gender split, and location alignment independently then averages. Each dimension is binary or range-overlap. Weight: 30%.

Engagement quality scorer
Critical

Normalizes influencer engagement rate against their follower tier benchmark. Above-benchmark scores high. Penalizes inflated follower counts. Weight: 25%.

Campaign history scorer
Critical

Checks past collaboration categories for niche overlap. Verified match = positive boost, no history = neutral, mismatch pattern = small penalty. Weight: 10%.

Weighted score aggregator
Critical

Combines all four sub-scores into a final 0–100 match score. Rounds to whole number. Stores result and all sub-scores in the matches table.

Score breakdown output
High

Returns not just the headline score but each sub-score labelled and explained. Exposed via API so the frontend can render the full breakdown bar.

PHASE 3 — FASTAPI BACKEND

Search & filter endpoint
Critical

Accepts niche, location, follower range, min engagement threshold. Returns a JSON list of influencer profiles sorted by match score for the requesting brand.

Profile detail endpoint
Critical

Returns full influencer record including demographics, content style, past campaigns, and pre-computed match score breakdown for the current brand session.

Match generation endpoint
Critical

Triggered when a brand runs a search. Computes or retrieves match scores for all returned influencers. Stores new scores to avoid redundant computation.

Session authentication
High

Token-based auth so brand and influencer accounts see only their own data. Lightweight — no OAuth needed at MVP stage. Tokens expire after session end.

Input validation layer
High

FastAPI Pydantic models validate all incoming request bodies and query parameters. Returns structured 422 errors for malformed inputs — no silent failures.

API documentation (auto)
Medium

FastAPI auto-generates /docs (Swagger UI) and /redoc pages. Used internally for testing endpoints and can be shown in course demonstrations.

PHASE 4 — STREAMLIT FRONTEND

Marketplace search page
Critical

Filter sidebar with niche dropdown, location input, follower range slider, and min engagement slider. Results render as cards sorted by match score in real time.

Influencer profile card
Critical

Shows name, niche badge, follower count, engagement rate, location, and headline match score as a percentage. Clicking opens the full profile detail view.

Full profile detail view
Critical

Expanded view with audience demographics, content style tags, past collaboration list, and the full match score breakdown visualized as a labelled bar chart.

Match score breakdown bar
High

Visual bar split into four colored segments (niche, audience, engagement, history) proportional to each sub-score. Makes the algorithm transparent and readable.

Brand input form
High

Simple sidebar form where a brand enters their niche, target audience, and budget. Used to personalize match scores before running a search.

Influencer profile builder
Medium

Form for influencers to enter their niche, follower count, engagement rate, audience demographics, content style, and past collaborations. Writes to the database.

Sort & rank controls
Medium

Toggle between sorting by overall match score, by engagement rate only, or by follower count. Lets brands prioritize differently depending on campaign goal.

Analytics summary panel
Low

A simple summary showing total influencers in the database, average engagement rate by niche, and distribution of match scores for the current search. Adds data richness.

PHASE 5 — TESTING & VALIDATION

Algorithm correctness tests
Critical

Unit tests for each sub-scorer. Exact niche match must return 100. Cross-niche mismatch must return below 30. Edge cases: missing demographics, zero history.

Ranking sanity checks
Critical

End-to-end test: fitness brand search must return fitness influencers above gaming influencers. Food brand must not surface tech creators in top 5 results.

API endpoint tests
High

Test all FastAPI endpoints with valid and invalid inputs. Verify correct HTTP status codes, JSON structure, and that filter parameters correctly narrow results.

UI render tests
Medium

Verify that profile cards render correctly with all fields, that the score breakdown bar reflects actual sub-scores, and that filter changes update results correctly.

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